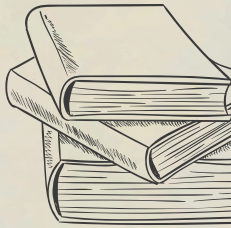
Personal Knowledge Graph Memory Engine and Career Development Assistant

Turning conversations into structured long-term memory for personalized and evolving career guidance



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01
Introduction

Motivation

The "Memory Gap" in Conversational AI

- Loss of depth and consistency across long-term interactions.
- Critical career signals are flushed out of the context window.

Limitations of Pure Vector RAG

- Flat similarity ignores complex relational context.
- Fails to capture the "Why" behind professional growth.

A "Brain + Expert" Architecture

Relational Memory (The "Brain")

Building persistent and evolving user personas through structured Knowledge Graphs.

Semantic Alignment

Seamlessly mapping unstructured conversational signals to industry-standard O*NET datasets.

Actionable Guidance (The "Expert")

Transforming memory into personalized gap analysis and professional development roadmaps.

02
Memory Architecture

Raw Conversation Data
ShareGPT-style multi-turn chats

↓

Cleaning & Splitting
Turn-levei records, filter noisy data

↓

Memory Extraction
Detect signals & extract memories

Memory Extraction

LLM Extraction (Groq/Llama-3.1):

Parsing unstructured dialogue into 10+ structured signal types (Skills, Tools, Goals, Work Styles, etc.).

Rule-Based Fallback:

Ensuring system reliability through deterministic keyword and pattern matching.

Knowledge Graph Storage

Relational Storage:

Utilizing NetworkX (MultiDiGraph) to preserve multi-hop connections between users, projects, and goals.

Deduplication & Linking:

Using fuzzy string matching (SequenceMatcher) to merge conversational variants into canonical nodes.

Memory Dynamics & Retrieval

Dual-Weighting Retrieval:

- Mention Frequency: Calculating node importance based on frequent mentions and project connectivity.
- Temporal Recency: Automatic decay of outdated nodes used for time-sensitive memory ranking.

Contextual Subgraph Surfacing:

- Retrieving semantically relevant clusters rather than isolated keywords to maintain conversational context.

Robustness of the Memory Engine

Structural Fidelity ((Validated via Gold Standard)):

- **0.97 Relation F1**
  Demonstrating exceptional accuracy in preserving complex relational links and multi-turn conversational context.

- **0.88 Entities F1**
  Enabling high-precision extraction of career-critical nodes across diverse professional domains.

Internal Stability:

- **1.0 Noise Robustness**
  Maintaining perfect resilience and anti-interference against typos, conversational fillers, and colloquial speech patterns.

- **1.0 Self-Consistency**
  Ensuring deterministic and stable graph construction results across redundant and independent processing runs.

03
Career Recommendation Architecture

Overview

Conversations between user and AI → Signal extraction (skills / interests / projects /...) → O*NET profile alignment → Two-stage recommendation → Gap analysis

Industry Alignment & Semantic Mapping

Each occupation

- Knowledge
- Skills
- Work Styles

User Embeddings

- Sentence Transformers (all-MiniLM-L6-v2)
- semantic similarity mapping

Cognitive Inference Engine - Inferring Implicit Traits

LLM extracts explicit signals & System infers implicit abilities

Examples:

mentioned many technical projects	→	Complex Problem Solving
repeated debugging	→	Attention to Detail
mentioned heavy workload	→	Stress Tolerance

Two-Stage Career Recommendation

Stage 1 — Vector Retrieval

- weighted cosine similarity
- Top-20 candidates

Stage 2 — LLM Re-ranking

- Groq + full context
- Top-5 + reasoning

Actionable Feedback: Gap Analysis

Compare User profile vs Target occupation

Output:

- missing skills
- missing knowledge
- development directions

Example:

- CS → + math / stats / finance → Quant

04
Evaluation for Career Recommendation Engine

Evaluation Framework & Results

AI-as-a-Judge

We leverage GPT-4o as an 'authoritative judge' to provide automated, expert-level evaluation for our Groq-extracted career data, overcoming the lack of ground-truth labels through high-reasoning semantic analysis.

- Qualitative scoring: Completeness, Faithfulness, and Career Utility
- Narrative-based profile validation: Generating a "Career Profile Narrative" to ensure the Agent's understanding matches user intent

Results

GPT-4o Career Judge
- Sample size: 30
- Completeness: 3.133 / 5
- Faithfulness: 3.3 / 5
- Career utility: 3.133 / 5

Results

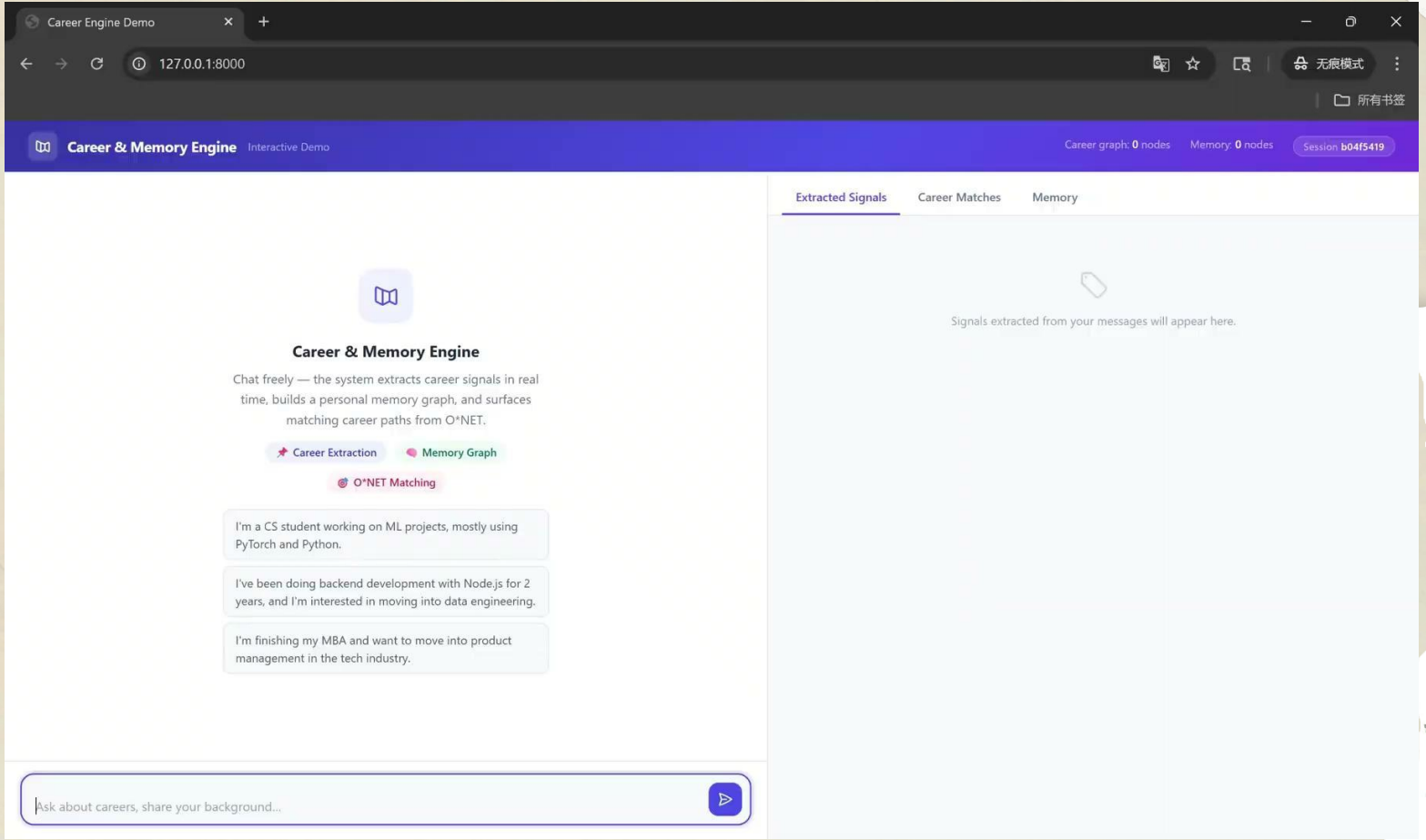
Narrative-based profile validation examples:

"dialogue_id": "career_dev_198",
"completeness": 3,
"faithfulness": 2,
"career_utility": 3,
"rationale": "Some key signals like **interest in portfolio optimization** and f**ast-paced work style** are missing or misrepresented, and some extracted items are **not grounded in the conversation**."

"dialogue_id": "career_dev_145",
"completeness": 4,
"faithfulness": 5,
"career_utility": 4,
"rationale": "The profile **captures most key signals from the conversation**, is well-grounded in the text, and provides **actionable insights**, though it could include more specific career path suggestions."

Demo

Demo


Career Engine Demo
127.0.0.1:8000
无痕模式
所有书签
Career & Memory Engine
Interactive Demo
Career graph: 0 nodes
Memory: 0 nodes
Session b04f5419
Extracted Signals
Career Matches
Memory
Career & Memory Engine
Chat freely — the system extracts career signals in real time, builds a personal memory graph, and surfaces matching career paths from O*NET.
Career Extraction
Memory Graph
O*NET Matching
I'm a CS student working on ML projects, mostly using PyTorch and Python.
I've been doing backend development with Node.js for 2 years, and I'm interested in moving into data engineering.
I'm finishing my MBA and want to move into product management in the tech industry.
Signals extracted from your messages will appear here.
Ask about careers, share your background...

05
Conclusion

Key Takeaways

The Power of Structured Memory

- Bridging Objective Data (O*NET) and Subjective Reasoning (LLM).
- Creating a coherent, evolving persona from unstructured dialogue.

Key Project Impact

- Beyond simple Q&A: Persistent decision support for career growth.
- High Precision: 0.97 Relation F1 score and 4.31 Faithfulness.

Future Vision

- Dynamic Embedding alignment and Interactive Graph Visualization.
- Human-in-the-loop: User-centric memory refinement.

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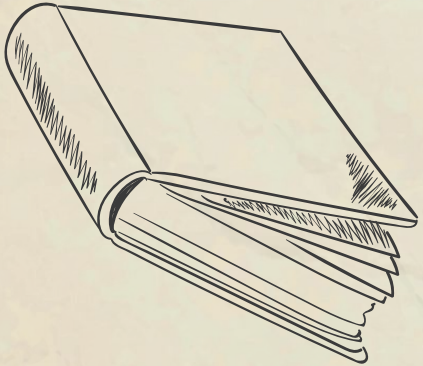

Thanks!